

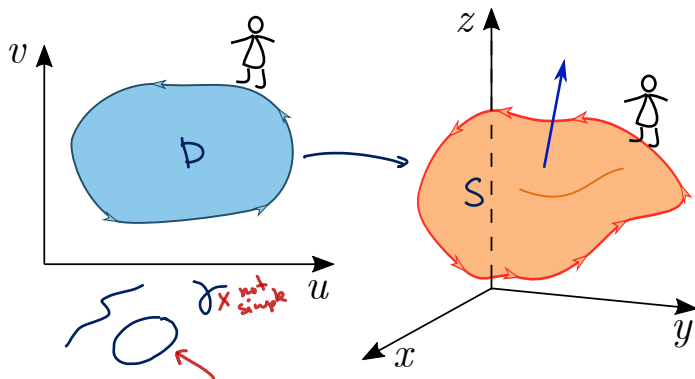
16 Stokes' and Gauss' Divergence Theorems

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Boundary curve



The boundary of D is a simple closed oriented curve, which is mapped to the boundary of the surface S via $\mathbf{r}$. We will use the convention that the positive orientation of the normal of S is such that walking along the curve, in the positive direction, on the side of the surface that corresponds to this positive normal, the surface enclosed by the curve is on the left.

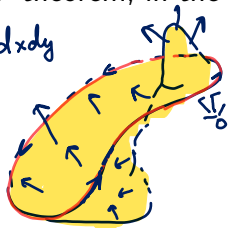
Theorem 16.1 (Stokes)

Let S be an oriented surface with parametrisation $\mathbf{r}$ that is one-to-one on a two-dimensional region D . Let C be the positively oriented simple closed boundary curve of S that is also piecewise smooth and let $\mathbf{F}$ be a continuously differentiable vector field on S . Then

$$\iint_S (\underbrace{\nabla \times \mathbf{F}}_{\text{curl } \mathbf{F}}) \cdot d\mathbf{S} = \oint_C \mathbf{F} \cdot d\mathbf{s}.$$

Green's theorem is a special case of Stokes' theorem, in the case when the surface S lies in the xy -plane.

$$\iint_D \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dx dy$$



conservative $\tilde{\mathbf{F}}$:

$$\tilde{\mathbf{F}} = \nabla \phi$$

$$\Rightarrow \nabla \times \tilde{\mathbf{F}} = \mathbf{0}$$

Also for a conservative field $\mathbf{F}$ we have $\oint_C \mathbf{F} \cdot d\mathbf{s} = \iint_S \mathbf{0} \cdot d\mathbf{S} = 0$.

Example 16.2

Evaluate $\oint_C \mathbf{F} \cdot d\mathbf{s}$ where $\mathbf{F}(x, y, z) = (y^2, x, z^2)$ and C is the curve of intersection of the plane $y + z = 2$ and the cylinder $x^2 + y^2 = 1$ (orient C to be counter clockwise when viewed from above).

$$\oint_C \mathbf{F} \cdot d\mathbf{s} = \iint_S \nabla \times \mathbf{F} \cdot d\mathbf{S} = \iint_D \underbrace{\begin{pmatrix} 0 \\ 0 \\ 1-2y \end{pmatrix}}_{(1-2y)} \cdot \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} dx dy \quad \textcircled{=}$$

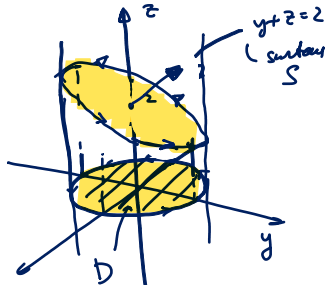
Stokes' thm $\uparrow$
S

$$\nabla \times \mathbf{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & x & z^2 \end{vmatrix} = 0\hat{i} + 0\hat{j} + (1-2y)\hat{k}$$

$$\textcircled{=} \int_0^{2\pi} \int_0^1 (1-2r\sin\theta) \cdot r \, dr d\theta = 2\pi \cdot \frac{1}{2} = \pi$$

$$\mathbf{T}_x \times \mathbf{T}_y = \begin{pmatrix} -\frac{\partial y}{\partial x} \\ -\frac{\partial x}{\partial y} \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

parametrisation: $\mathbf{r}(x, y) = x\hat{i} + y\hat{j} + (2-y)\hat{k}$
correct orientation!



Example 16.3

Use two methods to calculate $\iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} dS$, where S is the surface $z = x^2 + y^2$, $0 \leq z \leq 4$ with $\hat{\mathbf{n}}$ directed downward and $\mathbf{F} = (\underline{y}, \underline{z}, \underline{x^2 y^2})$.

$0 \leq z \leq 4$ with $\hat{\mathbf{n}}$ directed downward and $\mathbf{F} = (\underline{y}, \underline{z}, \underline{x^2 y^2})$.

Using Stokes' thm:

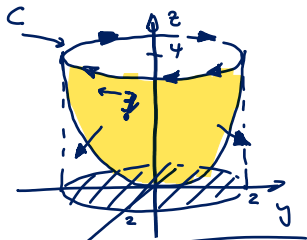
$$I = \oint_C \mathbf{F} \cdot d\mathbf{s} = \int_a^b \mathbf{F}(\underline{c}(t)) \cdot \underline{c}'(t) dt \quad \ominus$$

$$\ominus \int_0^{2\pi} \begin{pmatrix} -2\sin t \\ 4 \\ * \end{pmatrix} \cdot \begin{pmatrix} -2\sin t \\ -2\cos t \\ 0 \end{pmatrix} dt$$

$$= \int_0^{2\pi} \underbrace{4\sin^2 t}_{\text{correctly oriented}} - \underbrace{8\cos t}_{\text{correctly oriented}} dt = \dots = 4\pi$$

$$\underline{c}(t) = \begin{pmatrix} 2\cos t \\ -2\sin t \\ 4 \end{pmatrix}$$

$$\underline{c}'(t) = \begin{pmatrix} -2\sin t \\ -2\cos t \\ 0 \end{pmatrix}$$



$$\begin{aligned} & \left. \begin{aligned} z &= x^2 + y^2 \\ z &= 4 \end{aligned} \right\} C \\ & x^2 + y^2 = 4 \\ & x = 2\cos t \quad t \in [0, 2\pi] \\ & y = -2\sin t \quad \text{correctly oriented} \\ & z = 4 \\ & \Rightarrow \text{parametric eqns of } C \end{aligned}$$

Direct:

$$\nabla \times \underline{F} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y & z & x^2 y^2 \end{vmatrix} = \begin{pmatrix} 2x^2 y - 1 \\ -2xy^2 \\ -1 \end{pmatrix}$$

$$I = \iint_{x^2+y^2 \leq 4} \begin{pmatrix} 2x^2 y - 1 \\ -2xy^2 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 2x \\ 2y \\ -1 \end{pmatrix} dx dy \quad \textcircled{=}$$

$$\textcircled{=} \iint_{x^2+y^2 \leq 4} 4x^3 y - 2x - 4xy^3 + 1 \, dx dy = \underline{4\pi}$$

Surface S is parametrised by $\underline{r}(x,y) = x\underline{i} + y\underline{j} + \underline{(x^2+y^2)k}$

$$(x,y) \in D,$$

$$D = \{(x,y) \mid x^2 + y^2 \leq 4\}$$

$$\underline{T}_x \times \underline{T}_y = \begin{pmatrix} \cancel{\partial f / \partial x} \\ -\cancel{2x} / \partial y \\ 1 \end{pmatrix} = \begin{pmatrix} -2x \\ -2y \\ 1 \end{pmatrix}$$

↑
reverse orientation

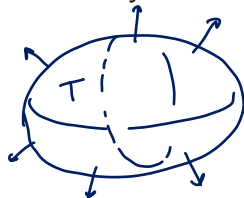
Theorem 16.4 (Gauss' Divergence Theorem) solid: $\overline{T} = \text{int } T \cup S$

Let T be a bounded solid simple region in the three-dimensional space, such that its boundary S can be represented as the finite union of parametrised non-overlapping surfaces with outward normals. Let $\mathbf{F}$ be a vector field which is continuously differentiable on T . Then

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_T \text{div } \mathbf{F} \, dV.$$

Gree's flux is equal to

$$\oint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, ds = \iiint_T \text{div } \mathbf{F} \, dx \, dy \, dz$$



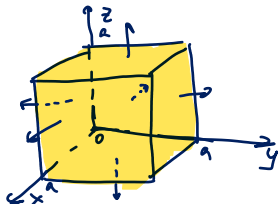
In other words, the outward flux of $\mathbf{F}$ across the surface S equals the triple integral of $\text{div } \mathbf{F}$ over T .

Note that here by 'simple' region we mean a region that is z -, y - or x -simple region, with the relevant simple shadow Ω , so that the triple integral is well-defined. 'Solid' means that the region is genuinely three-dimensional: its closure coincides with the closure of its interior.

Example 16.5

Evaluate

$$\iint_S \mathbf{F} \cdot d\mathbf{S},$$



where $\mathbf{F}(x, y, z) = (x^2, y^2, z^2)$ and S is the surface of the cube $0 \leq x \leq a$, $0 \leq y \leq a$, $0 \leq z \leq a$, with normals pointing outwards.

Divergence theorem.

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iiint_T \operatorname{div} \mathbf{F} \, dV = \int_0^a \int_0^a \int_0^a (2x + 2y + 2z) \, dx \, dy \, dz \\ &= \underbrace{\int_0^a \int_0^a \int_0^a 2x \, dx \, dy \, dz}_{\text{same value}} + \underbrace{\int_0^a \int_0^a \int_0^a 2y \, dy \, dz \, dx}_{\text{same value}} + \underbrace{\int_0^a \int_0^a \int_0^a 2z \, dz \, dx \, dy}_{\text{same value}} = 3 \int_0^a \int_0^a \int_0^a 2x \, dx \, dy \, dz \\ &= 3 \cdot a^4 \end{aligned}$$

Example 16.6

Find the outward flux of the vector field $\mathbf{F}(x, y, z) = (z, y, x)$ through the unit sphere $x^2 + y^2 + z^2 = 1$.

Example 16.7

Suppose S is the surface of the volume bounded by $x^2 + y^2 = 4$, $z = 1$ and $z = 3$. Find the flux of the vector field $\mathbf{F} = (x^2y, yz, x^2z)$ by two different methods.

$$\iiint_T \operatorname{div} \mathbf{F} \, dV = \int_0^{2\pi} \int_0^2 \int_1^3 \operatorname{div} \mathbf{F} \, dz \, dr \, d\theta = \dots = 24\pi$$

$$\iint_{S_1} \mathbf{F} \cdot d\mathbf{S} + \iint_{S_2} \mathbf{F} \cdot d\mathbf{S} + \iint_{S_3} \mathbf{F} \cdot d\mathbf{S} = \dots = 24\pi$$

$12\pi \quad -4\pi \quad 16\pi$

